

Site2Cloud (S2C)

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Solutions Engineering

Agenda

- Site2Cloud Overview
- Site2Cloud Use Cases
  1. High Speed DC Connectivity with Backup VPN
  2. Shared Services Multi-Tenant Architecture (aka SaaS Provider)
  3. Overlapping IP Space Scenarios
- Other Services to Connect to External Networks
- SD-WAN Integration

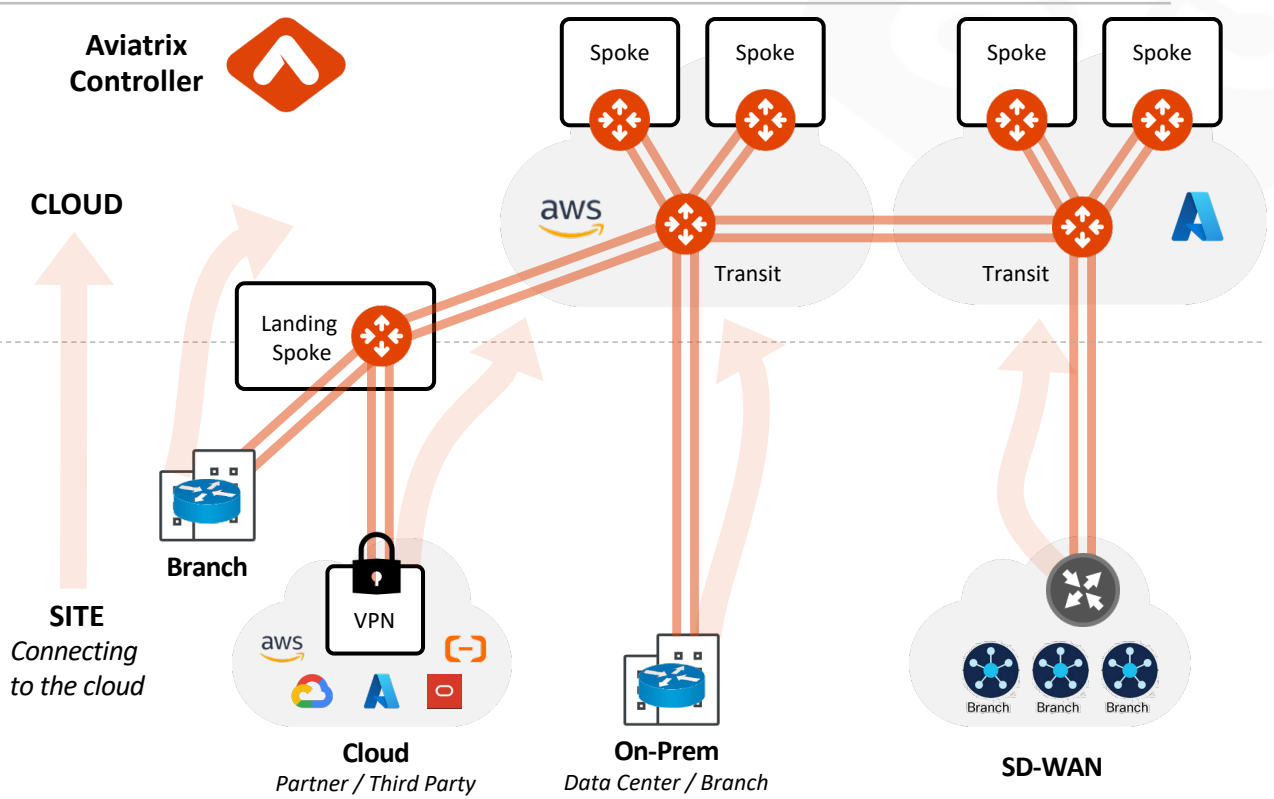
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Overview

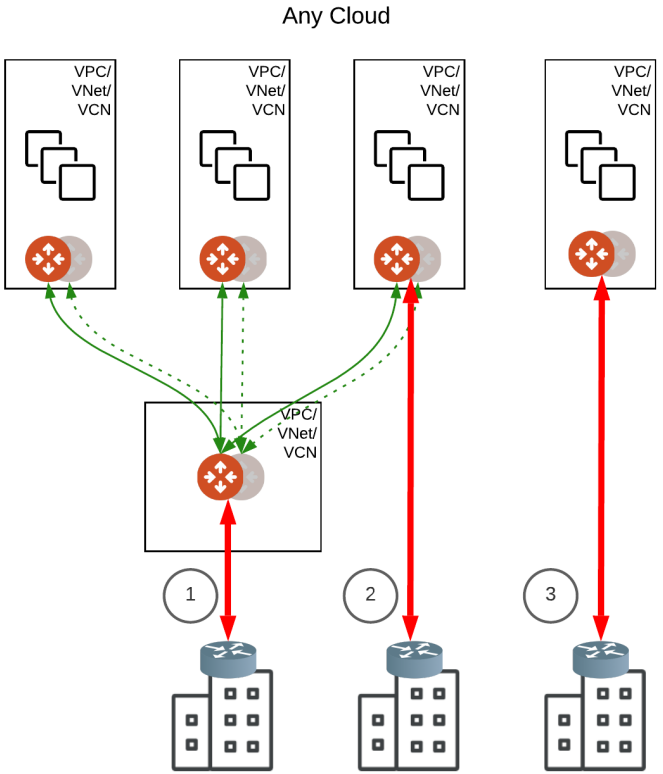
What is Site2Cloud?

- Connection from Public Cloud to:
  - On-Prem DC
  - 3rd Party Appliances, SD-WAN
  - Branch
  - Clouds Native Constructs (VPCs/VNets/VCNs)



Site2Cloud Landing Options

1. Transit Gateway
2. Spoke Gateway
3. “Standalone” Gateway (with HA Gateway)



Aviatrix at Remote Site

- CloudN hardware appliance
  - 1 RU physical device
  - Documented at https://docs.aviatrix.com/HowTos/old_cloudn_faq.html
- Aviatrix Edge virtual appliance (introduced in 6.7)
  - Virtual form factor
  - Supports OVA for ESXi
  - Supports KVM for Equinix
  - Supports Zero Touch Provisioning (ZTP) via cloud-init as well as ISO
  - Feature parity with CloudN
  - Documented at https://docs.aviatrix.com/HowTos/edge-2.0-workflow.html

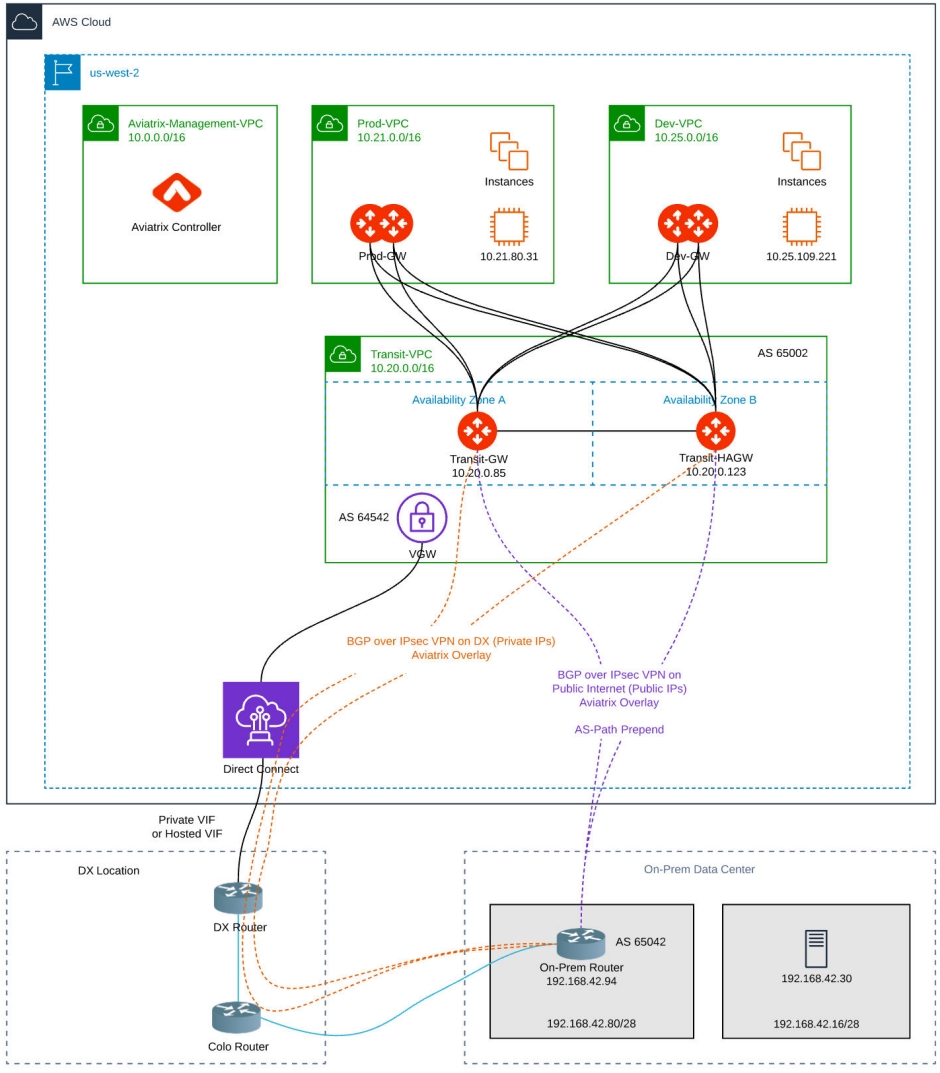


Use Cases

High Speed DC Connectivity with Backup VPN

High Speed DC Connectivity with Backup VPN

- Connecting on-prem data centers to the cloud via route-based Site2Cloud + BGP control plane, landing on Transit gateways
- Primary Site2Cloud is using private IPs to leverage the DX underlay
- Backup Site2Cloud is using public IPs to use the public Internet as underlay
- On both connections, ECMP can be enabled for Active/Active high performance or disabled (typically if on-prem has stateful firewalls)
- On-prem router is performing AS-path prepend on VPN routes advertised to Aviatrix transit over the VPN connection, to force Transit gateways to send traffic via the DX connection
- Additionally, on-prem router could use Weight or Local Pref, etc., to send traffic to the DX connection
- If DX connection goes down, traffic would automatically failover to Backup connection
- Branch connectivity is following a similar BGP-based Site2Cloud to Transit gateways, but it is typically only via VPN over the public Internet



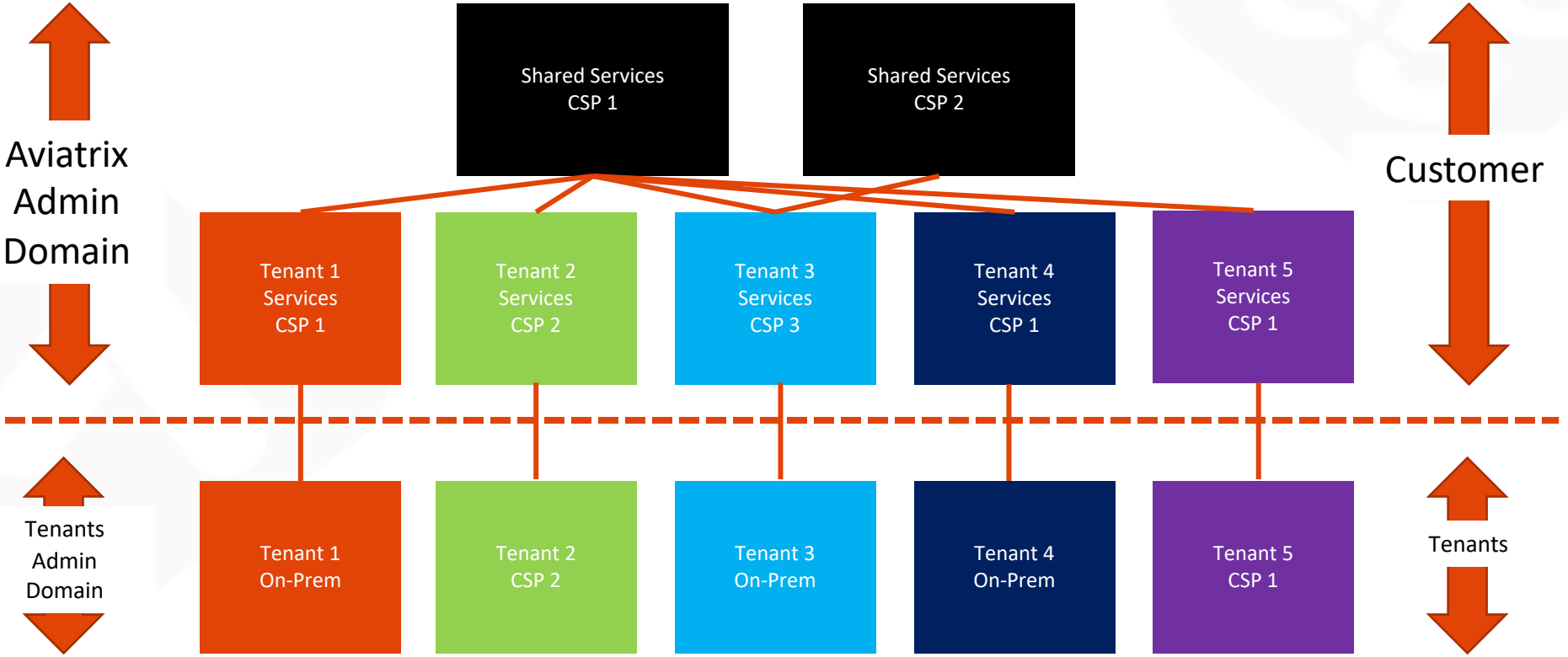


Use Cases

Shared Services Multi-Tenant Architecture

(aka Saas Provider)

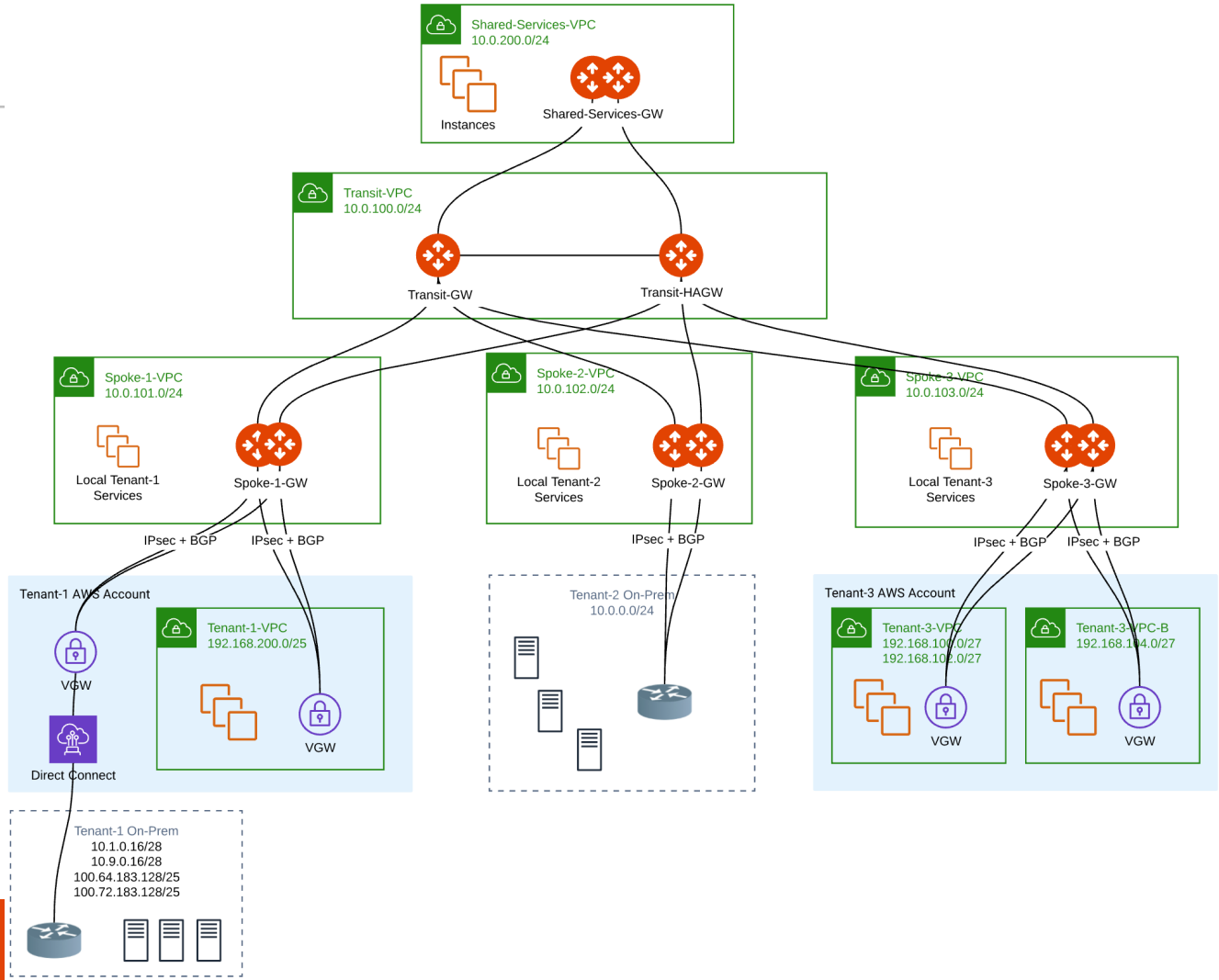
SaaS High-Level Architecture



Requirements and Solution

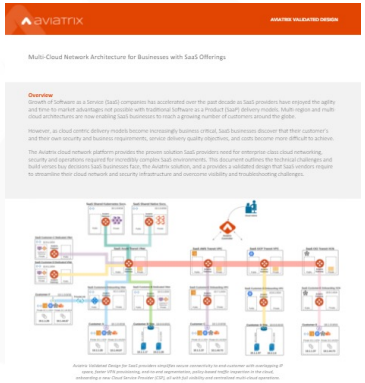
1. Connect a large number of tenants (1000+)
   - Distribute the tenants across Spoke gateways, for horizontal scaling and blast radius minimization
2. Provide both dedicated tenant services, and shared services
   - Host shared services in common Spoke VPCs
   - Host dedicated services in tenant-specific Spoke VPCs
3. Onboard the tenants with BGP: dynamic control plane that fits their operational model
   - Terminate BGP on the tenant Spoke gateways
4. Handle overlapping IPs across tenants, and between tenants and shared services
   - Use NAT on the tenant Spoke gateways
5. Maintain isolation across tenants
   - Use network domains (segmentation) on the tenant Spoke gateways
6. Provide the highest throughput to tenant services
   - Horizontal scaling
   - Tenant services are directly hosted in the Spoke VPC where BGP terminates
   - They're directly accessed by tenants, without the Transit layer being a bottleneck

Typical Architecture



Segmentation and NAT Support

- SaaS Providers Aviatrix Validated Design https://aviatrix.com/resources/design-guides/aviatrix-validated-design-saas-providers-infrastructure



Segmentation Domains

- Shared-Services
- Tenant-1
- Tenant-2
- Tenant-3

SNAT on Spoke1-GW for Tenant1-initiated traffic towards Shared Services:
- SRC CIDR = 192.168.200.0/25
- DST CIDR = 10.0.200.0/24
- SNAT IP = 10.0.101.10

SNAT on Spoke3-GW for Tenant3-initiated traffic towards Shared Services:
- SRC CIDR = 192.168.200.0/25
- DST CIDR = 10.0.200.0/24
- SNAT IP = 10.0.103.10

Shared-Services-VPC
10.0.200.0/24
Instances
Shared-Services-GW

Transit-VPC
10.0.100.0/24
Transit-GW
Transit-HAGW

Spoke-1-VPC
10.0.101.0/24
Local Tenant-1 Services
Spoke-1-GW
10.0.101.1

Spoke-2-VPC
10.0.102.0/24
Local Tenant-2 Services
Spoke-2-GW
10.0.102.10

Spoke-3-VPC
10.0.103.0/24
Local Tenant-3 Services
Spoke-3-GW
10.0.103.10

IPsec + BGP
IPsec + BGP
IPsec + BGP
IPsec + BGP
IPsec + BGP

Tenant-1 AWS Account
VGW
Direct Connect
Tenant-1-VPC
192.168.200.0/25
VGW

Tenant-2 On-Prem
10.0.0.0/24

Tenant-3 AWS Account
Tenant-3-VPC
192.168.100.0/27
192.168.100.0/27
VGW
Tenant-3-VPC-B
192.168.200.0/25
VGW

Tenant-1 On-Prem
10.1.0.16/28
10.9.0.16/28
100.64.183.128/25
100.72.183.128/25

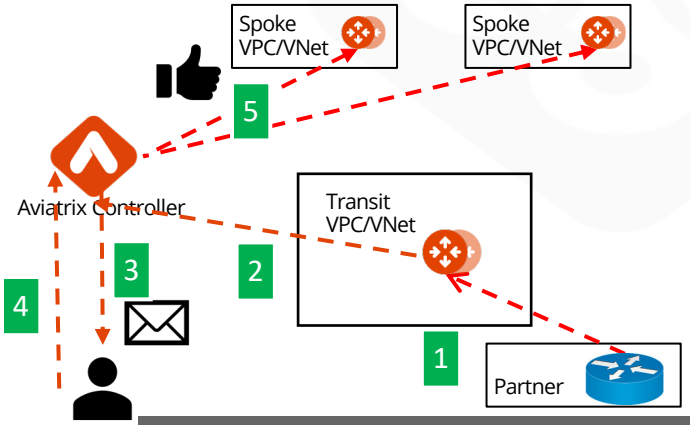
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BGP Route Approval

- Can explicitly approve any BGP-learned route from Partner or on-prem into the cloud network
- Prevents unwanted advertisement of routes such as 0/0 from Partner

1. New routes arrive at Transit Gateway
2. Transit Gateway reports new routes to Controller
3. Controller notifies admin via email
4. Admin logs in to Controller to approve
5. If approved, Controller programs the new routes to Spoke VPCs

- **Note:**
  - Route Approval completely blocks a BGP prefix to even be considered by control plane
  - Prefixes blocked are not even programmed in the Gateway route table



From Aviatrix Controller: Route Approval Request

NR no-reply@aviatrix.com
To

We removed extra line breaks from this message.

Time Detected: 2022-03-01 17:27:36.709787

Request approval for new learned CIDR(s):

Gateway: transit, Connection: onprem_to_transit-1, CIDRs(1): 0.0.0.0/0

To approve, please login to the Aviatrix Controller and go to Multi-Cloud Transit-> Approval.

Controller IP:
Controller Name: AWS Controller 6.6
Controller Version: UserConnect-6.6.5224 Time Detected: 2022-03-01 17:27:36.710310

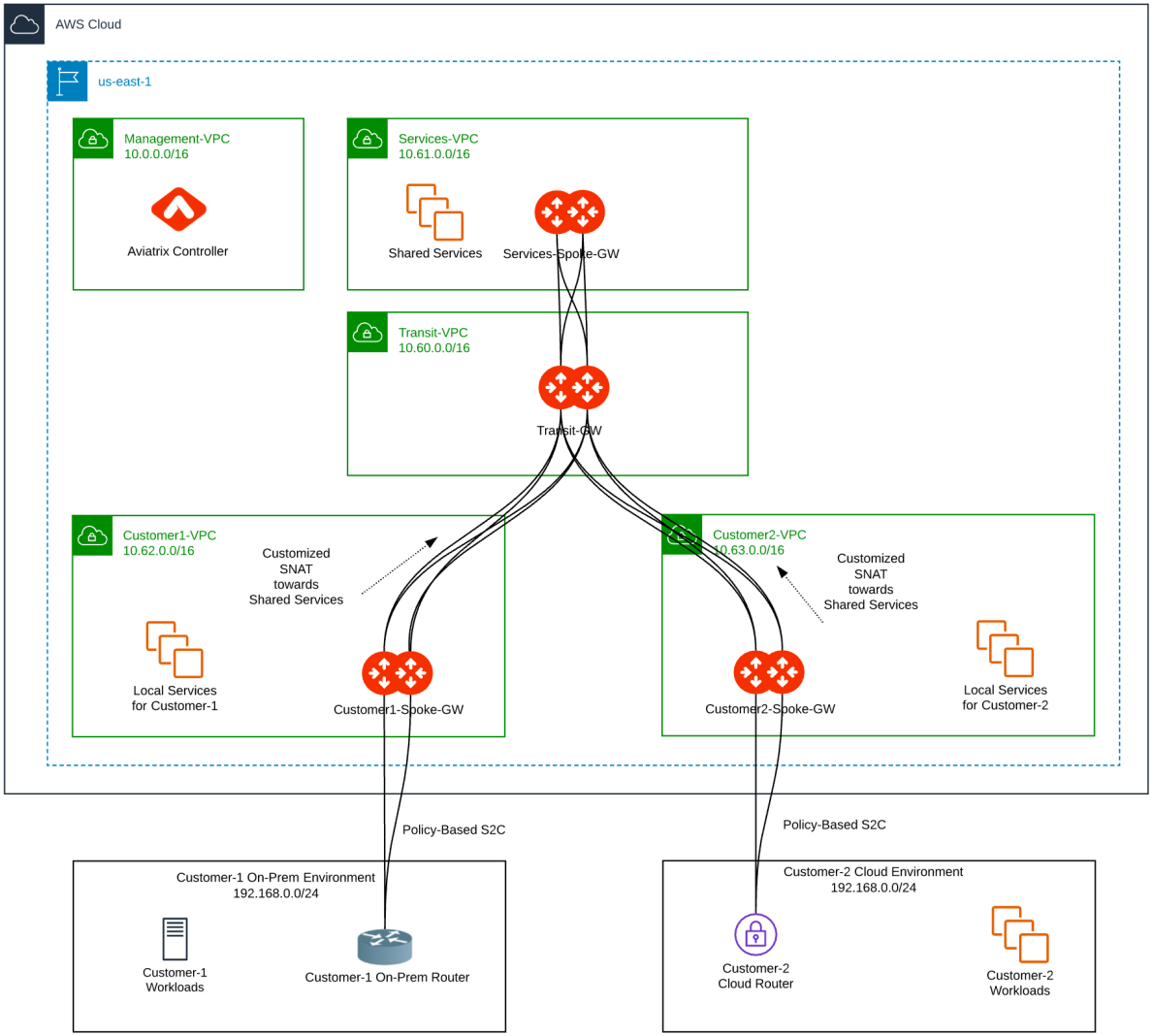


Use Cases

Overlapping IP Space Scenarios

SaaS Provider Scenario

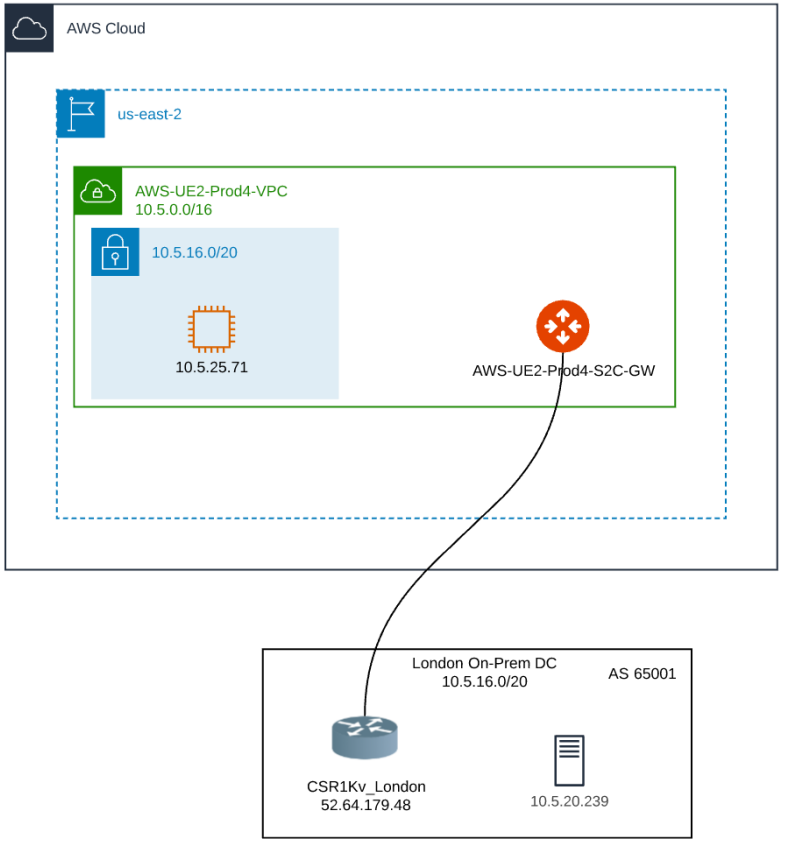
- Tenants could be on-prem or in their own cloud environment, separate from the customer cloud environment
- Tenants are onboarded via policy-based or route-based Site2Cloud with static routing, landing on ActiveMesh spoke gateways
- They land in their own VPCs to handle overlapping IP scenarios and provide them local services
- Customized SNAT is used to uniquely differentiate incoming overlapping tenant traffic when communicating with shared services



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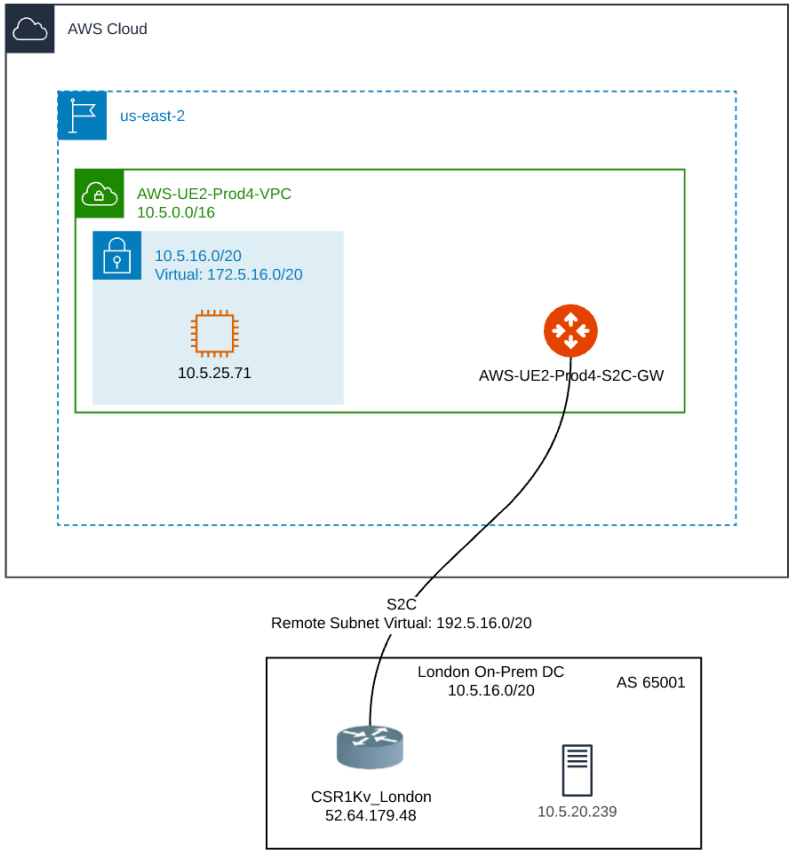
Requirements

- Need to connect overlapping networks between the cloud and on-prem
- Don’t want the on-prem router to implement any NAT
  - Keep it simple with no on-prem dependency
  - Many on-prem routers have no NAT, or very limited NAT
- The host information must be preserved
  - No NAT overload requirement anywhere
- The configuration must be simple and scalable



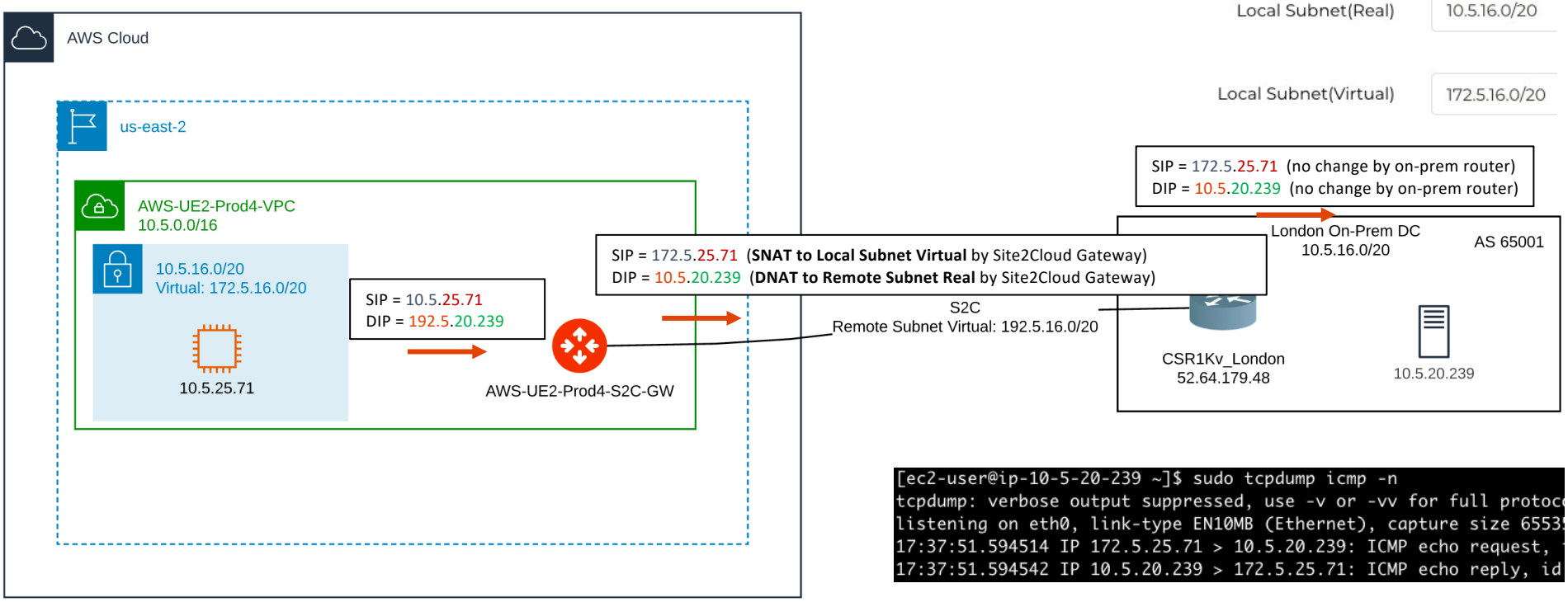
Solution – Mapped NAT with Route-Based Site2Cloud

- Virtual subnets, which are defined to be unique (not necessarily RFC1918), are used for communication between overlapping VPC and on-prem
- The Site2Cloud Gateway NATs between real subnets and virtual subnets
- There is no need for any on-prem NAT operations
- The configuration is extremely simple, and it does not require individual /32 NAT rules
- Route-based IPsec is required



Packet Walk

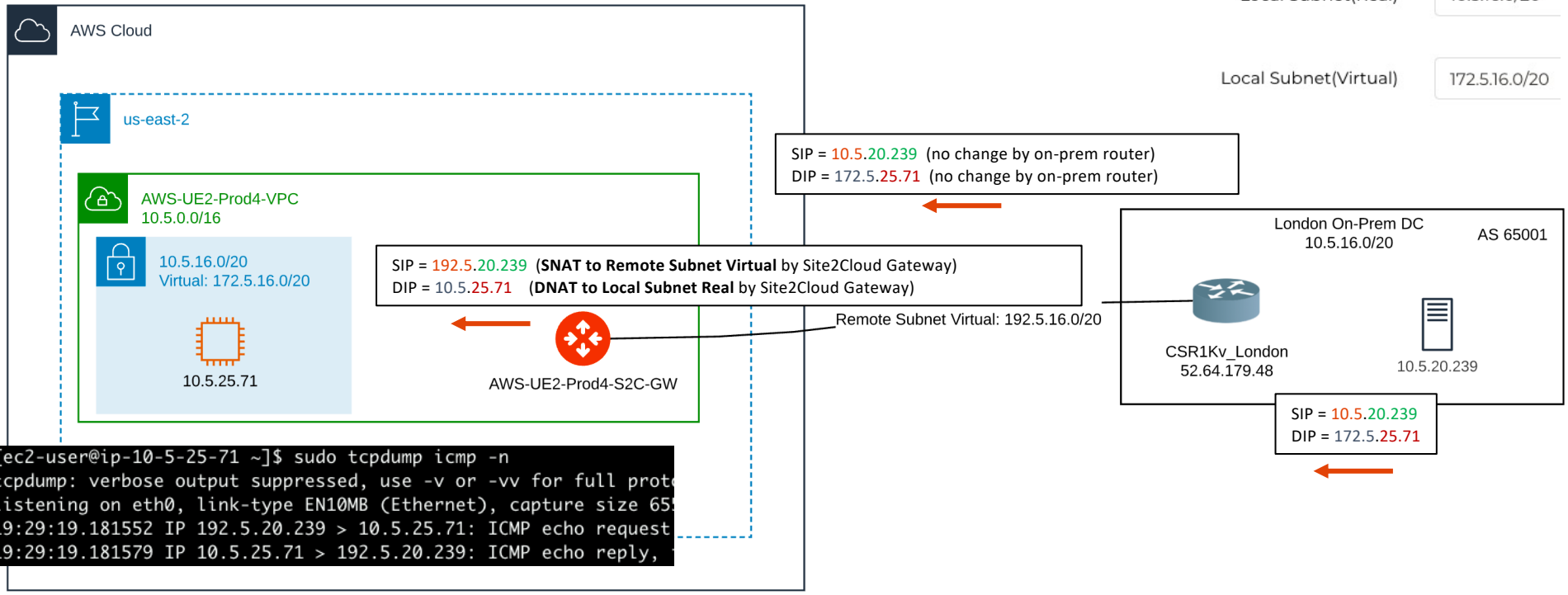
Remote Subnet (Real)	10.5.16.0/20
Remote Subnet (Virtual)	192.5.16.0/20
Local Subnet(Real)	10.5.16.0/20
Local Subnet(Virtual)	172.5.16.0/20




```
[ec2-user@ip-10-5-20-239 ~]$ sudo tcpdump icmp -n
tcpdump: verbose output suppressed, use -v or -vv for full protoco
listening on eth0, link-type EN10MB (Ethernet), capture size 6553
17:37:51.594514 IP 172.5.25.71 > 10.5.20.239: ICMP echo request,
17:37:51.594542 IP 10.5.20.239 > 172.5.25.71: ICMP echo reply, id
```

Packet Walk – Return Traffic

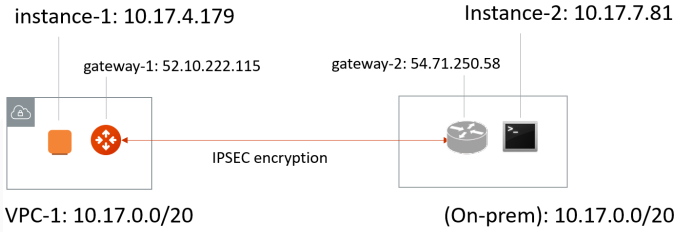
Remote Subnet (Real)	10.5.16.0/20
Remote Subnet (Virtual)	192.5.16.0/20
Local Subnet(Real)	10.5.16.0/20
Local Subnet(Virtual)	172.5.16.0/20



Overlapping IP Use Cases for Site2Cloud – Reference

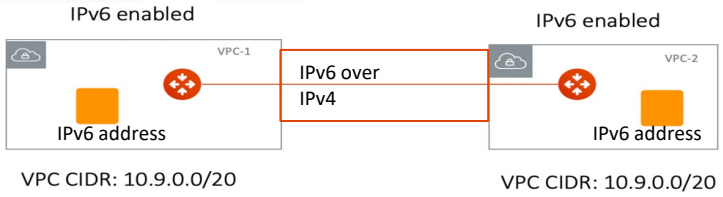
Overlapping CIDRs Within Cloud(s) or Directly to a VPC

https://docs.aviatrix.com/HowTos/connect_overlap_cidrs.html



Use IPv6 to Connect Overlapping VPC CIDRs

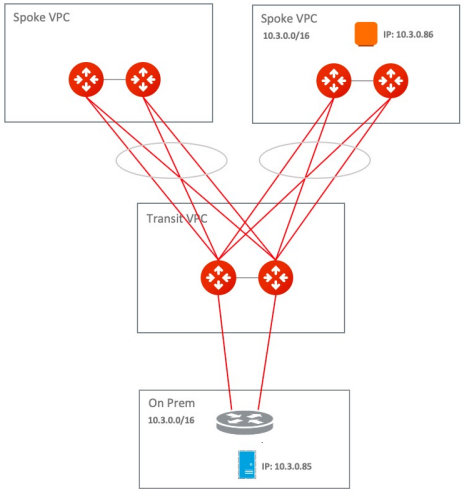
https://docs.aviatrix.com/HowTos/ipv6_peering.html



Requirement: EC2 instances can communicate with each other with IPv6

Aviatrix ActiveMesh with Customized SNAT and DNAT on Spoke Gateway

https://docs.aviatrix.com/HowTos/transit_solution_activemesh_spoke_snat_dnat_rfc1918.html



Other Overlapping IP CIDR Solutions

https://docs.aviatrix.com/HowTos/overlapping_network_solutions.html



Aviatrix Edge

Introducing Aviatrix Edge

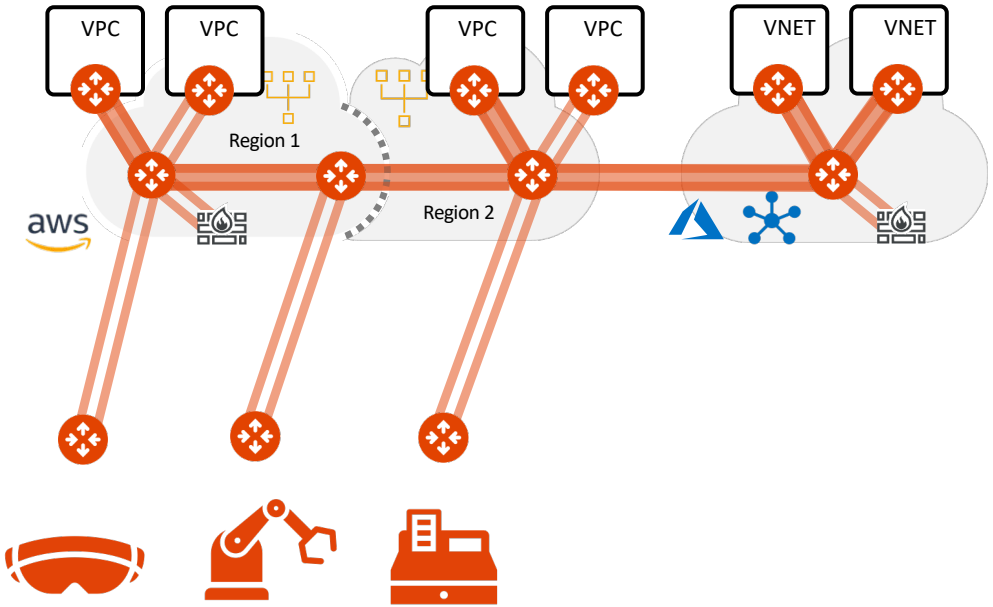
The only multi-cloud native platform with enterprise-grade visibility and control for public cloud and the edge
Aviatrix software in multiple form factors providing consistent network, security, and visibility to the edge.
Edge locations appear and behave as another VPC/VNET with spoke and transit capabilities.

Cloud Out Architecture

Simplified Edge Management

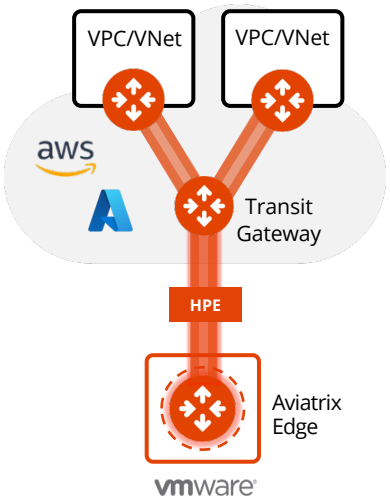
Consistent Secure Edge

Simplified Edge On-boarding

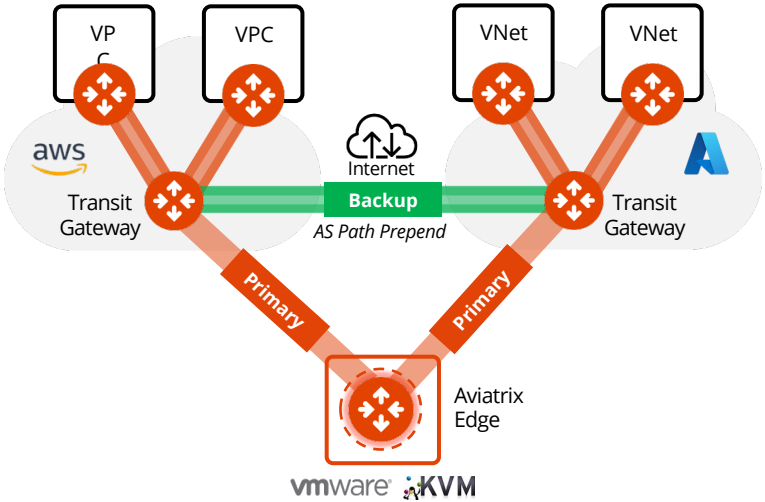


Aviatrix Edge Use Cases

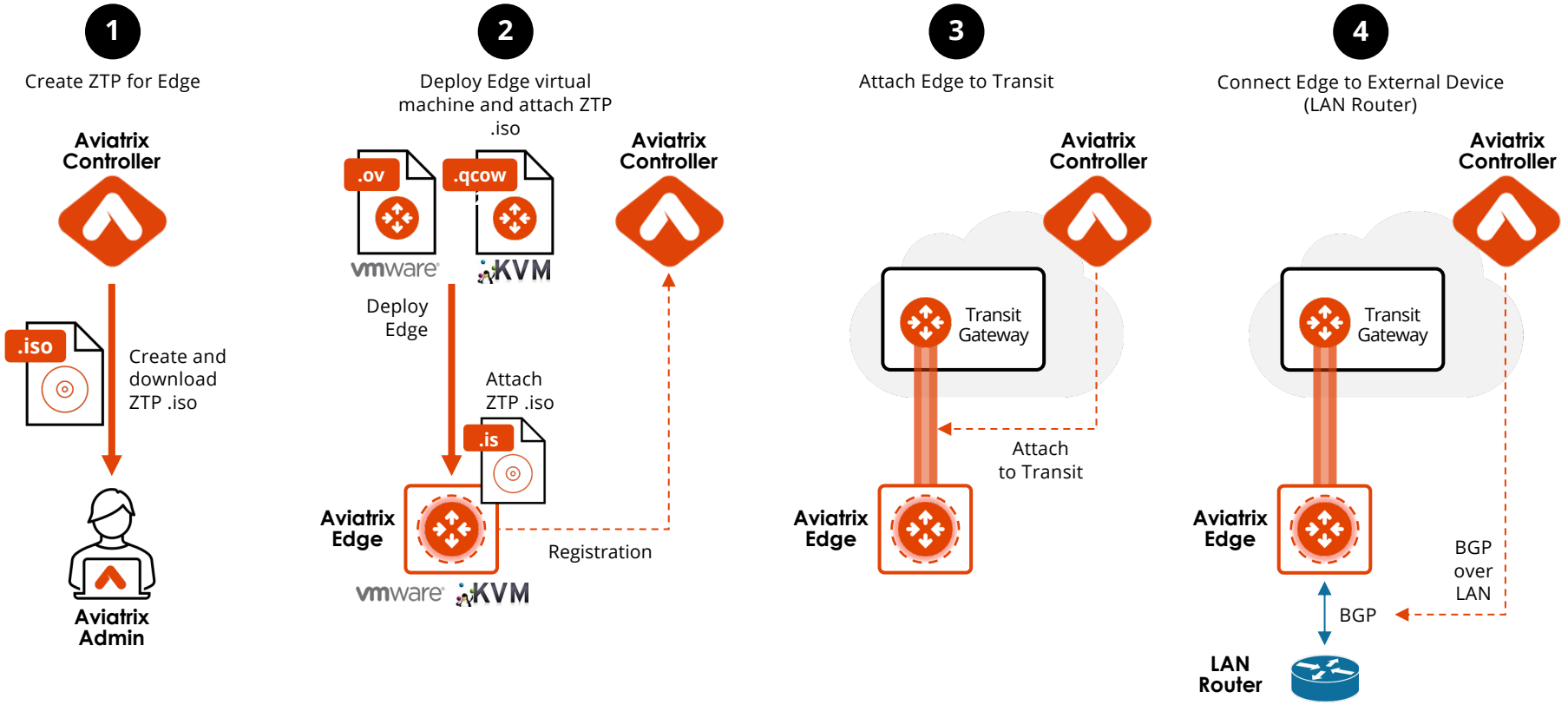
Extend the Aviatrix Platform to the Edge



Multi-Cloud Connectivity via Aviatrix Edge



Edge 2.0 Deployment Workflow - Demo

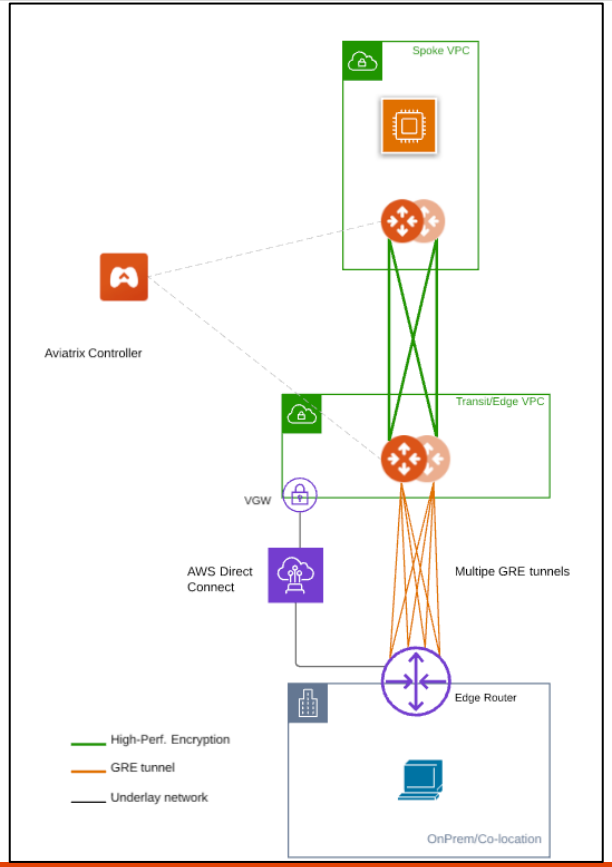


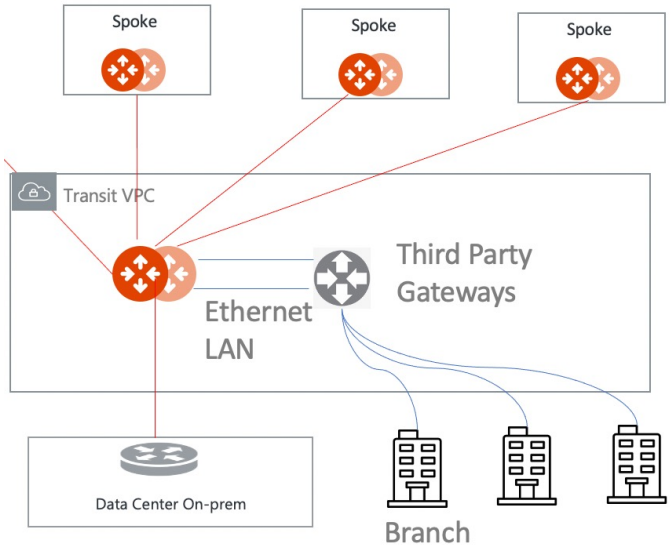


Other Services to Connect to External Networks

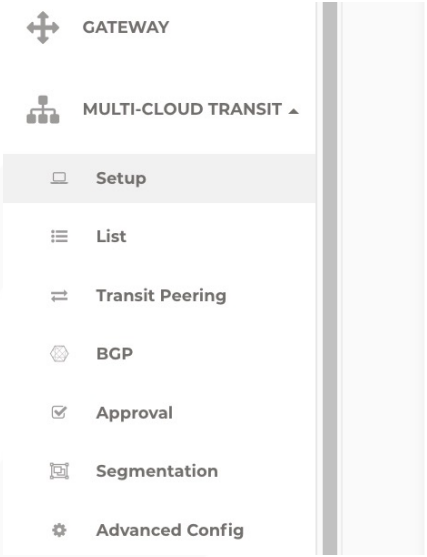
Connections to External Device

- IPsec (discussed already)
- BGP over GRE (AWS only)
  - Extends Aviatrix overlay to external networks without encryption, and without IPsec speed limitations
  - Useful for AWS DX
- BGP over LAN
  - Route exchange without any tunneling protocol
  - High-performance, widely compatible SD-WAN integration
  - Integrates with GCP Network Connectivity Center (NCC)





Configuration – MULTI-CLOUD TRANSIT > Setup > Attach tab > Step 3



External Connection

2 Connect to VGW / External Device / Aviatrix CloudN / Azure VNG Info

AWS VGW | External Device | Aviatrix CloudN | Azure VNG

BGP | Static Remote Route-Based

IPsec | GRE | LAN



SD-WAN Integration

Problem Statement

1. Limited cloud-native knowledge
2. Limited throughput of IPsec
3. Route summarization and traffic engineering
4. Challenges with security and service chaining









Solution – SD-WAN integration with Aviatrix

- BGP based integration with SD-WAN cloud instances
  - BGP over IPsec
  - BGP over LAN
  - BGP over GRE
- Service chaining by inspecting traffic with Next Gen Firewalls
- Advanced Traffic Engineering and Filtering options
- All other Aviatrix benefits apply

viptela
Viptela is now part of Cisco.
CISCO

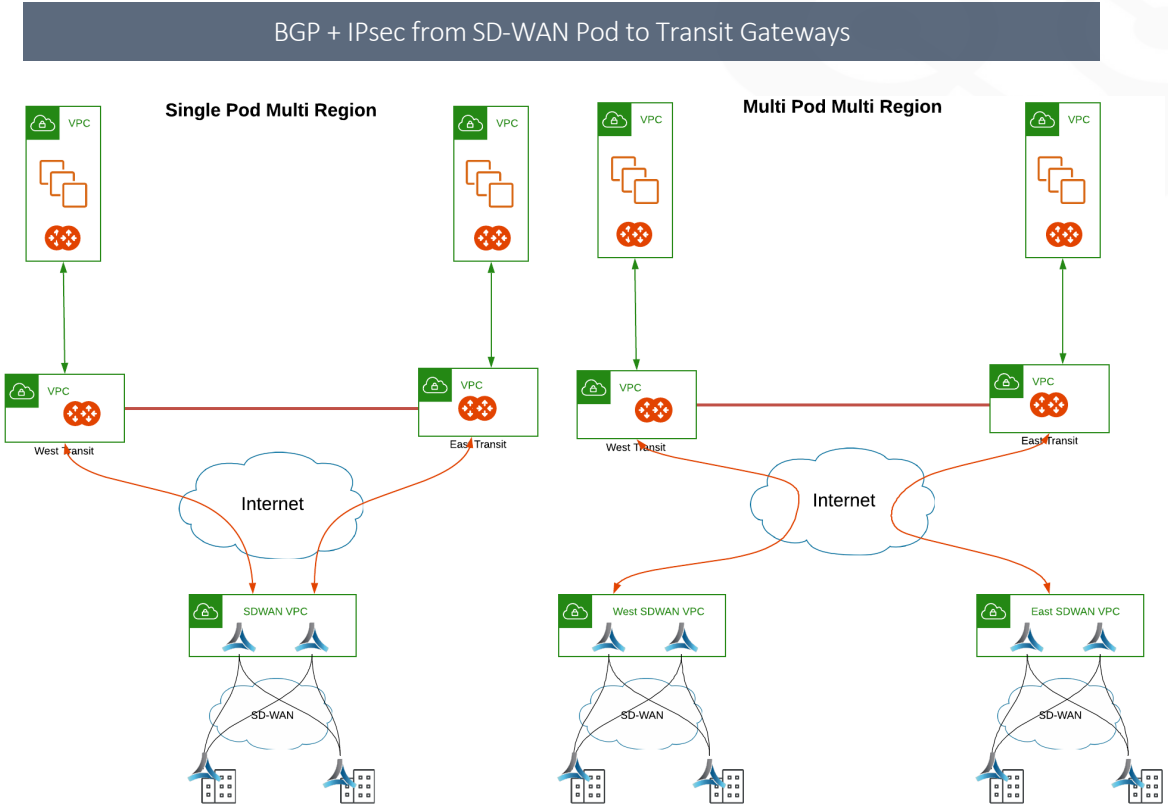
velocloud
vmware

silver peak
aruba

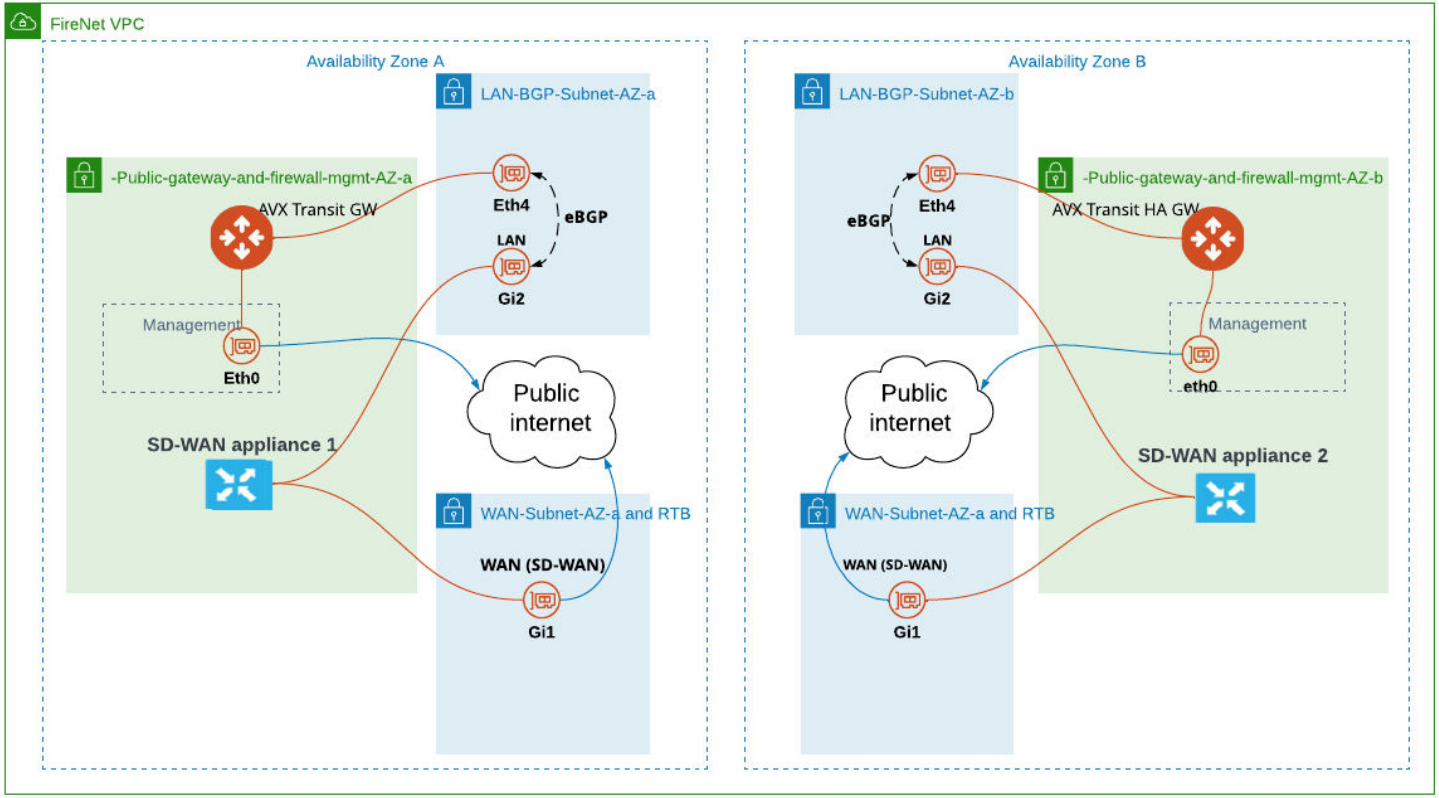
CISCO
Meraki

Uses BGP over LAN Feature

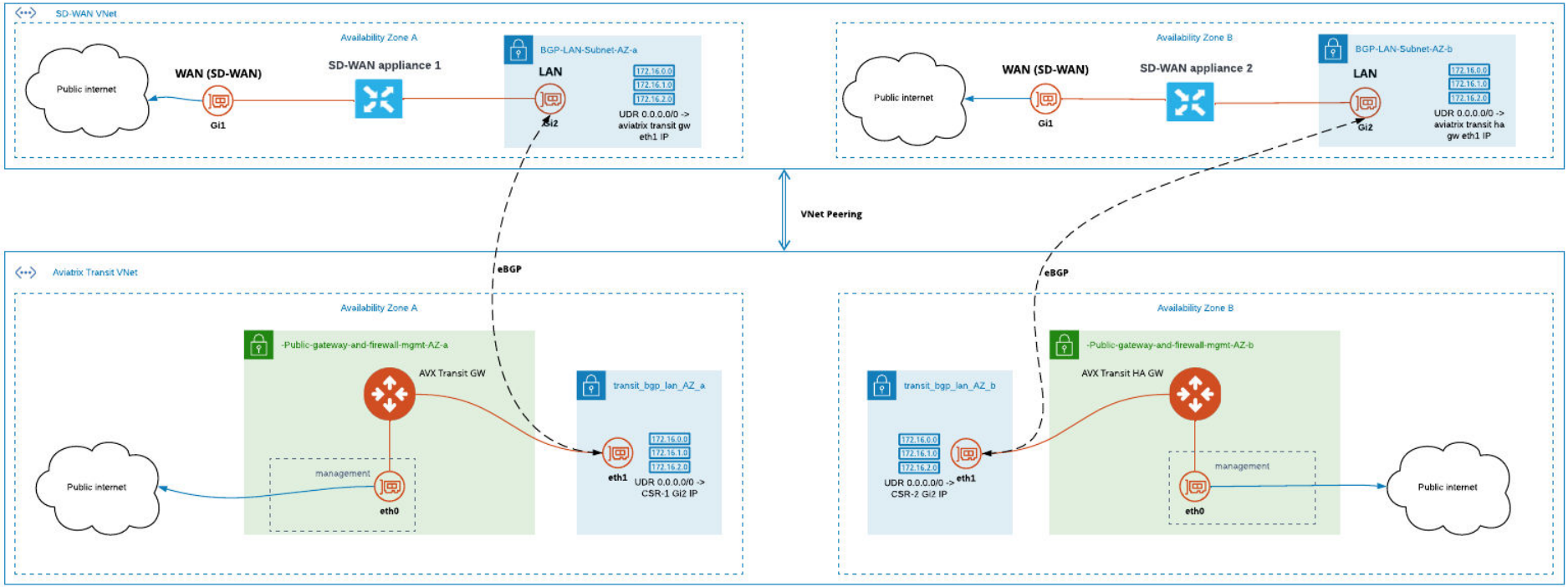
- Why BGP peering over an IPsec tunnel to a Transit Gateway might not be preferred
  - IPsec limits throughput
  - Not all vendors support BGP over IPsec (Meraki) for dynamic routing integration
  - Not all vendors/CSPs support GRE
- BGP over LAN supports BGP peering natively on Aviatrix Transit Gateway
- No tunneling required



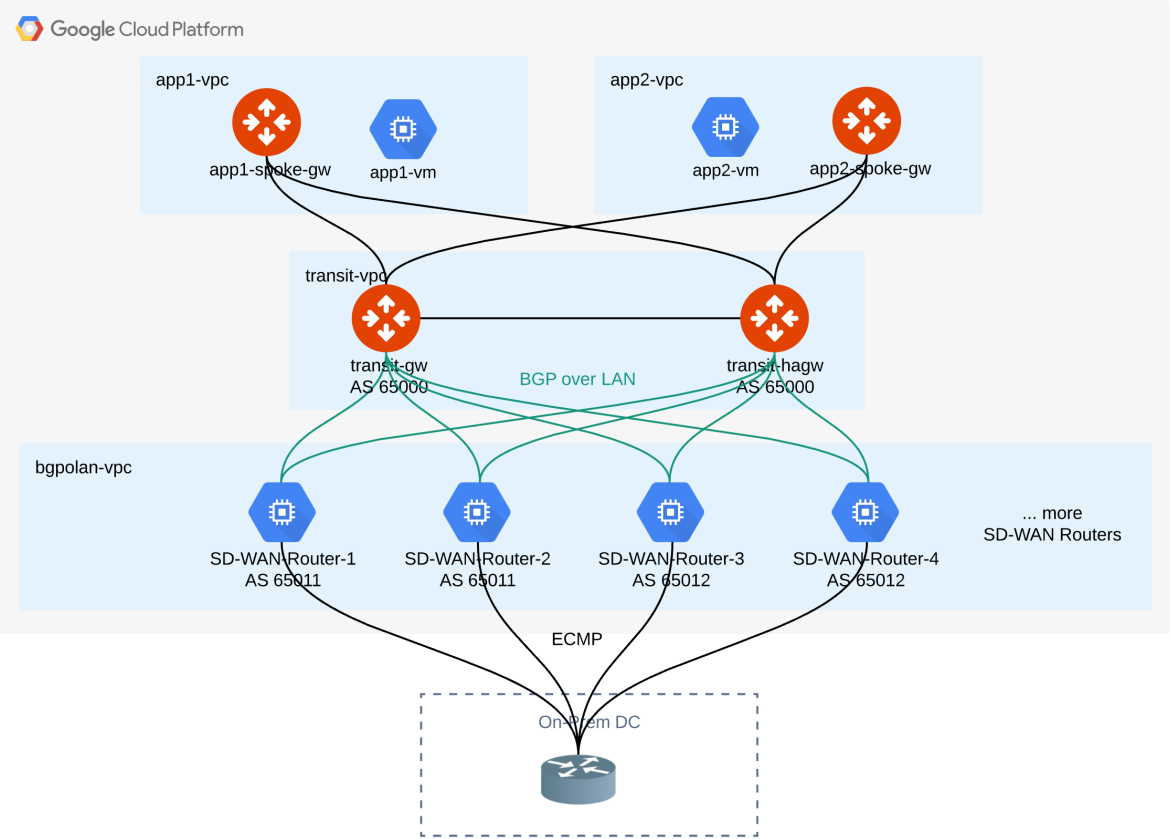
BGP over LAN in AWS



BGP over LAN in Azure



How does it work in GCP?



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Next: Lab 8